
EDUCATION

- **Indraprastha Institute of Information Technology** Delhi, India
Doctor of Philosophy (Ph.D.), Computer Science, CGPA: 8.24 *Aug. 2020 – Present*
Advisor: Dr. A. V. Subramanyam
Research Area: AI Safety
Coursework: Advanced Machine Learning, Deep Learning, Machine Learning, Probability and Random Processes, Convex Optimization, Reproducing Kernel Hilbert Spaces and Applications, Meta Learning, Digital Image Processing
- **Guru Gobind Singh Indraprastha University** Delhi, India
Bachelor of Technology in Computer Science and Engineering *Aug. 2013 – Jul. 2017*

EXPERIENCE

- **Microsoft** Bangalore, KA, India
Applied Sciences Intern *May 2026 – Jul. 2026*
- **National University of Singapore** Singapore
Visiting Scholar *May 2025 – Oct. 2025*
 - Worked on robustness and alignment of foundation models with focus on jailbreak simulation, and adversarial prompt defense mechanisms.
- **Indraprastha Institute of Information Technology** Delhi, India
Research Scholar *Aug. 2020 – Present*
 - Defending deep learning models against adversarial attacks on image classification using purification techniques with diffusion models.
 - Developed Language Guided Adversarial Purification method for adversarial robustness.(Patent filed)
 - Investigating adversarial robustness by combining foundation models from computer vision and natural language for image classification.*Teaching Assistant* *Aug. 2020 – May 2024*
 - Advances in Deep Learning [M'23]
 - Computer Vision [W'21]
 - Digital Image Processing [M'21, M'22]
 - Natural Language Processing [M'20]
 - Statistical Machine Learning [W'22, W'23, W'24, W'25]
- **Medway Technologies Pvt. Ltd.** Remote
Consultant *Dec. 2021 – Dec. 2022*
 - Developed an OCR pipeline to extract text from images of printed and handwritten medical prescriptions.
 - Compare extracted text with database to identify prescribed medicines and patient information.
- **Animaker Inc.** Bangalore, KA, India
Research Scientist (Machine Learning and Computer Vision) *Jul. 2017 – Aug. 2020*
 - Contributed in building patented text to video platform Steve AI
 - Developed and deployed a machine learning pipeline for background subtraction in images and videos.
 - Designed and implemented algorithms to identify and rank relevant sections of long videos based on activity detection, facial recognition, attention, and saliency analysis.
 - Built a recommendation system for layout optimization, enabling intelligent placement of characters and text in visual content.
 - Developed color recommendation models for scene components, utilizing user-preferred hue templates from large datasets and creating quantitative scoring models to assess the quality of color combinations.
 - Provided dynamic color palette suggestions based on existing scene components and background images or videos, enhancing visual coherence and aesthetic appeal.

PUBLICATIONS

- **H. Singh**, A. V. Subramanyam, Mitigating Distribution Shift for Adversarial Purification, *Under Review*.
- S. Rajput*, **H. Singh***, A. V. Subramanyam, Learning Adversarial Manifold for Jailbreaking at Scale, **Equal Contribution, Under Review*.
- **H. Singh**, Z. Xu, A. V. Subramanyam, M. Kankanhalli, Do Prompts Guarantee Safety? Mitigating Toxicity from LLM Generations through Subspace Intervention, *Under Review*.
- **H. Singh**, A. V. Subramanyam, S. Rajput, M. Kankanhalli, Nearest Neighbor Projection Removal Adversarial Training, *IEEE Transactions on Artificial Intelligence*, April 2026.
- **H. Singh**, A. V. Subramanyam, Language Guided Adversarial Purification, *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Seoul, Republic of Korea, 2024, pp. 7685-7689.

PATENTS

- A.V. Subramanyam, **H. Singh**, A method and a system for defending a deep learning model, *Indian Patent No. 592301. Granted, 15 Jun. 2026*.
- V.C. Bhagath, **H. Singh**, System and Method for Language Learning, *Indian Patent No. 567815. Granted, 24 Jun. 2025*.

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL
- **Databases:** MongoDB, MySQL
- **Libraries and Software Packages:** PyTorch, Hugging Face, OpenCV, Numpy, Pandas, Scikit-learn, Matlab

AWARDS AND ACHIEVEMENTS

- Dean's List - Best TA Award, Advances in Deep Learning
- Best Performance - Research award for outstanding performance at Animaker Inc
- Bagged 1st Position at "Syntaxomania" a coding event organized by the IEEE group at Amitech 2015
- Recipient of scholarship for the year 2011-2012 from Shanti Gyan Niketan Sr. Sec. School
- Recipient of Certificate of Merit from CBSE for securing 10 CGPA

CERTIFICATIONS AND TRAININGS

- Deep Learning Specialization by DeepLearning.ai on Coursera
- Machine Learning by Stanford University on Coursera
- Machine Learning Foundations: A Case Study Approach by University of Washington on Coursera
- Machine Learning: Regression by University of Washington on Coursera
- The Data Scientist's Toolbox by Johns Hopkins University on Coursera

COMMUNITY SERVICE

- Reviewer, IEEE MLSP 2025
- Reviewer, IEEE ICASSP 2025
- Volunteer, IEEE ICASSP 2024